



Betruck Ouma

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• EDUCATION & TRAINING

05/09/2022 - Current - NYERI, KENYA

BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING- DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

- Mechanics of machines & structural mechanics (statics, dynamics, kinematics and dynamic load analysis)
- Mechanical design & modeling (3D parametric modeling, reverse engineering, CAD/CAM integration)
- Simulation & engineering analysis (CFD, FEA, aerodynamic flow studies, motion analysis)
- Manufacturing & workshop technology (CNC & conventional machining)
- Applied mathematics & control systems (numerical analysis, control theory modeling, statistical DOE)
- Thermodynamics & fluid mechanics

15/01/2018 - 01/04/2022 - BUSIA, KENYA

KENYA CERTIFICATE OF SECONDARY EDUCATION (KCSE)- SIGALAME BOYS HIGH SCHOOL

09/09/2026

CERTIFIED SOLIDWORKS PROFESSIONAL (CSWP)- DASSAULT SYSTÈMES

- Advanced 3D modeling (complex parametric design, multi-body modeling)
- Assembly & configuration management (advanced assembly mates, collision detection, & design tables)
- Engineering analysis (applied mass property calculations and design intent optimization)

01/09/2026

CERTIFIED SOLIDWORKS ASSOCIATE (CSWA)- DASSAULT SYSTÈMES

- CAD fundamentals (basic parametric modeling techniques, and design intent)
- Engineering drafting (generating standard 2D drawings from 3D solid models)
- Part & assembly modeling (constructing fundamental mechanical parts & assemblies)

14/05/2024 - 13/09/2024

SOFTWARE AND WEB DEVELOPMENT- POWER LEARN PROJECT

- Application development (cross-platform application development utilizing Flutter and Dart, alongside Database Management using SQL)

02/09/2024 - 19/12/2024

ETHICAL HACKING- CISCO NETWORKING ACADEMY

- Cybersecurity & network defense (ethical hacking methodologies, network vulnerability assessments, and system diagnostics)

• WORK EXPERIENCE

09/02/2026 - 15/05/2026 - NAIROBI, KENYA

MECHANICAL ENGINEERING ATTACHÉNUMERICAL MACHINING COMPLEX (NMC)

- CNC machining & precision manufacturing (programming and operating multi-axis CNC machines)
- Metallurgy & metal casting (pattern design allowances, metal casting, quality control, hot-dip galvanizing)
- Fabrication & structural joining (advanced thermal joining including SMAW, MIG, and TIG welding)
- Reverse engineering by CAD integration for manufacturing

27/01/2025 - 18/04/2025 - NAIROBI, KENYA

MECHANICAL ENGINEERING ATTACHÉASSOCIATED MOTORS, ISUZU

- Vehicle diagnostics & electromechanics (electromechanical troubleshooting and system fault resolution)
- Automotive servicing & maintenance (periodic maintenance, servicing, and structural repairs)
- Quality assurance & safety compliance (executing Pre-Delivery Inspections & enforcing quality standards)

08/01/2024 - 29/03/2024 - NYERI, KENYA

MECHANICAL ENGINEERING ATTACHÉDEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

- Mechanical design & optimization (leveraging CAD/CAM tools to improve fabrication efficiency)
- Workshop operations & machining (lathe, milling, and drilling equipment for precision part production)
- Systems maintenance & fabrication (installation of dynamic mechanical systems)

● **SKILLS**

Ansys workbench CFX | SOLIDWORKS (CSWP) | SolidWorks (CSWA) | Statistical analysis (MINITAB) | AutoCad 2D -3D | MATLAB | Microsoft Office | Mechanical Cad Design

● **HONOURS AND AWARDS**

23/02/2026 Dedan Kimathi University of Technology

Top Academic Performer Recognized as the best student in the Bachelor of Science in Mechanical Engineering program for three consecutive academic years (1st Year, 2nd Year, and 3rd Year).

● **PROJECTS**

03/11/2025 - 04/02/2026

10kW Crossflow Turbine design Performed complete hydraulic dimensioning and structural design of a 10 kW crossflow turbine for deployment on Kenyan rivers, sizing the runner geometry, nozzle and casing.

31/08/2025 - 19/11/2025

Biogas Generator Conversion Converted a gasoline engine into a 4 kW biogas-fueled power generator, redesigning fuel delivery, compression ratio, and combustion chamber.

19/03/2026 - 05/05/2026

Re-Block Extruder Proposed an extruder machine converting recycled plastics and sand into paving blocks, targeting plastic waste reduction and low cost construction materials.

15/05/2024 - 01/08/2024

Maize Cob Crusher & Compactor Designed a maize cob crusher and compactor to process maize crop residue into a densified output, sizing both the crushing and compaction mechanisms for required throughput.

24/02/2025 - 01/04/2025

Speed Reducer Gearbox Designed a compound reverted helical gearbox (78.5 kW, 1200-155 rpm) via AGMA gear-stress and shaft fatigue analysis, achieving factor of safety (FOS) > 2 across all shafts.